

Critical Thinking Assignment 7: Labs Lessons 12

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IT315-2: Introduction to Networks

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Critical Thinking Assignment 6: Labs Lessons 12

This documentation is part of the Critical Thinking 7 Assignment from ITS315: Introduction to Networks at Colorado State University Global.

The Assignment Direction:

Module #6: uCertify Lab Simulations

For this assignment, you will complete multiple lab simulations. Activities include identifying network connection types, connecting networks to the internet, configuring routers, etc. You will take a screenshot upon completion of each lab and include the screenshots in the submitted assignment.

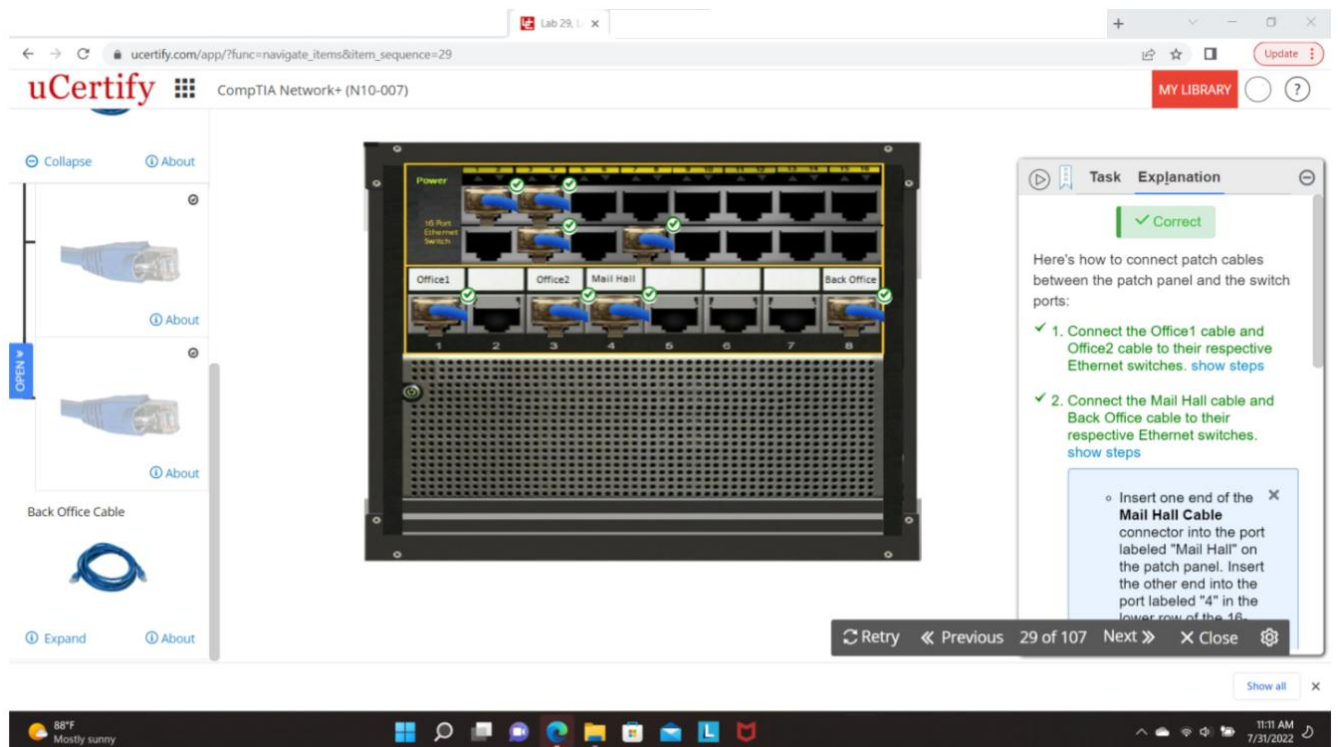
Access uCertify and login, go to Labs, and complete the tasks in the following lab simulations:

- 12.2.9 Identifying network attacks
- 12.3.7 Scanning using nmap
- 12.3.8 Running the Nessus vulnerability scan
- 12.5.3 Identifying types of firewall
- 12.6.1 Creating a remote access VPN connection

After completing the task, click Submit >> Evaluate >> Record my answer to record your answer. Take a screenshot of each of the labs and paste the screenshot into a Word document. The document should have a title page that includes your name, date, school name, section, course name, and instructor name.

Submit the assignment in Canvas.

Please ensure your screenshot includes your name, date, and timestamp as shown in the image below.



Screenshots

Figure 1
12.2.9 Identifying network attacks

The screenshot shows a web browser window displaying a uCertify quiz titled "12.2.9 Identifying network attacks". The quiz is presented in a matching format. On the left, under the heading "Attack", there is a list of five items: 1. Malware, 2. Man-in-the-middle, 3. EMI interception, 4. Dumpster Diving, and 5. FTP bounce. On the right, under the heading "Description", there are five options labeled A through E. Arrows indicate the correct matches: Malware to A, Man-in-the-middle to B, EMI interception to C, Dumpster Diving to D, and FTP bounce to E. Each description is marked with a green checkmark. The interface includes tabs for "Correct Answer", "Compare", and "Your Answer". A sidebar on the right shows the quiz progress and a list of categories, including "Network Security" and "Categories of Network Attacks". The bottom of the screen shows a Windows 10 taskbar with the time 4:30 PM on 2/2/2025.

Attack

1. Malware
2. Man-in-the-middle
3. EMI interception
4. Dumpster Diving
5. FTP bounce

Description

- A. An attacker copies information traveling over the wire
- B. An attacker sends a PORT command specifying the IP address of a third party
- C. An attacker gets in the direct path between a client and a server and eavesdrops on their conversation
- D. A software program designed to damage or take other unwanted actions on a computer system
- E. A technique used to retrieve information that could be used to carry out an attack on a network

Activity Explanation

✓ Correct

Types of network attacks are described below:

- ✓ **Malware:** A software program designed to damage or take other unwanted actions on a computer system
- ✓ **EMI interception:** An attacker copies information traveling over the wire
- ✓ **Man-in-the-middle:** An attacker gets in the direct path between a client and a server and eavesdrops on their conversation
- ✓ **FTP bounce:** An attacker sends a PORT command specifying the IP address of a third party
- ✓ **Dumpster diving:** A technique used to retrieve information that could be used to carry out an attack on a network

Lesson

Network Security

Categories of Network Attacks

Figure 2
12.3.7 Scanning using nmap

The screenshot shows a web browser window displaying a uCertify quiz titled "12.3.7 Scanning using nmap". The quiz is presented in a video format. The main content area shows a Windows 10 desktop with a blue background and a taskbar. A video player is embedded in the center, showing a play button and the text "Scanning using nmap". The interface includes tabs for "Activity", "Explanation", and "Answer". A sidebar on the right shows the quiz progress and a list of categories, including "Network Security" and "Defending Against Attacks". The bottom of the screen shows a Windows 10 taskbar with the time 4:34 PM on 2/2/2025.

Activity Explanation Answer

✓ Correct

Here's how to scan using nmap:

- ✓ Scan using nmap, show steps

Transcript

Lesson

Network Security

Defending Against Attacks

Figure 3

12.3.8 Running the Nessus vulnerability scan

The screenshot shows a uCertify quiz interface. The main area displays a Windows 10 desktop background. On the right, the quiz content is visible under the 'Explanation' tab. It shows a 'Correct' status and a list of steps to run a Nessus vulnerability scan:

- ✓ 1. Login to nessus. [show steps](#)
- ✓ 2. Run the vulnerability scan. [show steps](#)
- ✓ 3. Observe the scan results. [show steps](#)

Below the steps is a video player titled 'Running the Nessus vulnerability scan' with a play button icon. A 'Transcript' tab is also visible. The bottom of the interface shows a navigation bar with 'RESET', 'PREVIOUS', '90 of 107', 'NEXT', 'RETRY', 'SETTINGS', and 'CLOSE' buttons. The system clock at the bottom right indicates 4:44 PM on 2/2/2025.

Figure 4

12.5.3 Identifying types of firewall

The screenshot shows a uCertify quiz interface. The main area displays a matching exercise titled 'Identifying types of firewall'. It consists of two columns: 'Description' and 'Firewall type'. The descriptions are:

1. Inspects traffic to identify unique sessions
2. Inspects traffic leaving the inside network as it goes out to the Internet
3. Inspects traffic based solely on a packet's header
4. Filters traffic based on ACL-like rules; however, lacks flexibility

The firewall types are:

- A. Packet-filtering
- B. Stateful

The correct matches are indicated by arrows and green checkmarks:

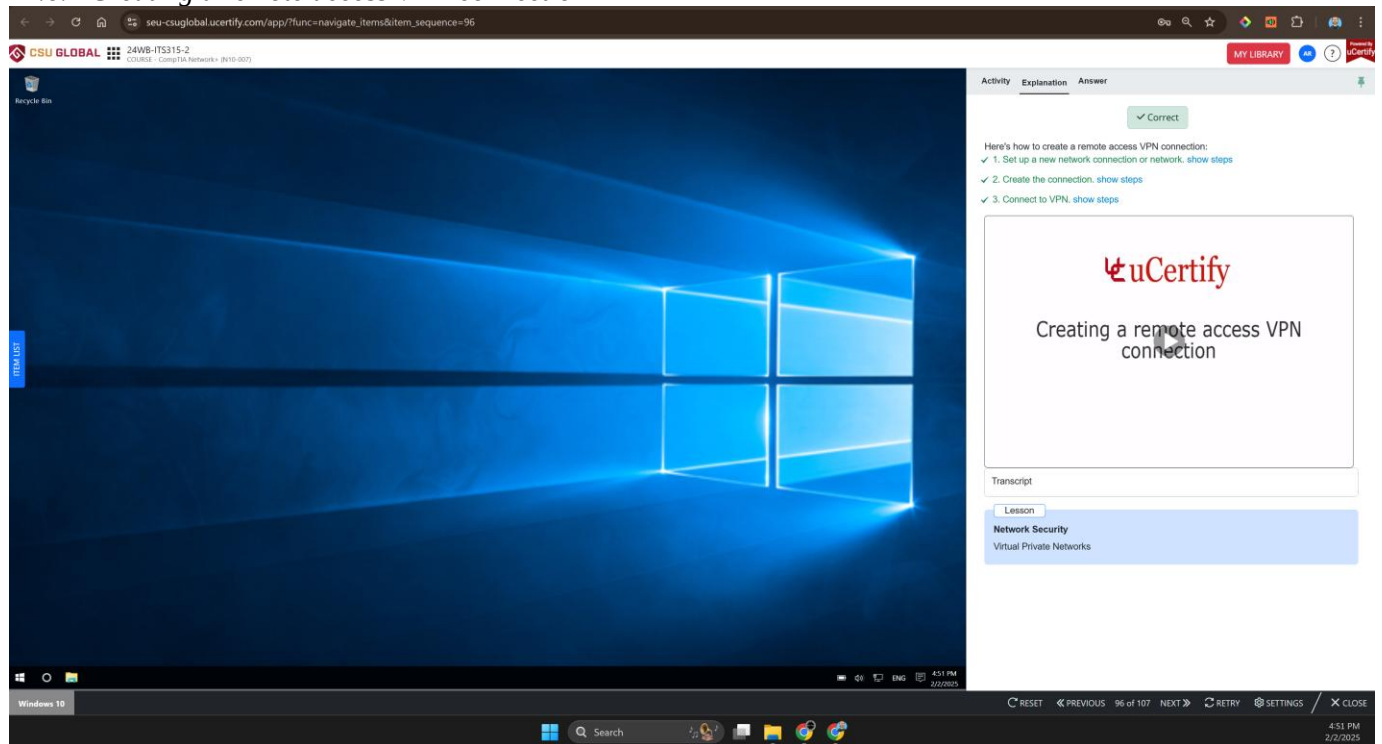
- 1. Inspects traffic to identify unique sessions → B. Stateful
- 2. Inspects traffic leaving the inside network as it goes out to the Internet → B. Stateful
- 3. Inspects traffic based solely on a packet's header → A. Packet-filtering
- 4. Filters traffic based on ACL-like rules; however, lacks flexibility → A. Packet-filtering

At the bottom, there are buttons for 'A. Packet-filtering' and 'B. Stateful'. The right side of the interface shows the 'Explanation' tab with a 'Correct' status and a list of firewall types with their descriptions:

- ✓ • **Packet-filtering firewall:** Inspects traffic based solely on a packet's header. It filters traffic based on ACL-like rules. However, a packet-filtering firewall lacks flexibility.
- ✓ • **Stateful firewall:** Inspects traffic leaving the inside network as it goes out to the Internet. The process of inspecting traffic to identify unique sessions is called stateful inspection.

The bottom navigation bar includes 'RESET', 'PREVIOUS', '92 of 107', 'NEXT', 'RETRY', 'SETTINGS', and 'CLOSE' buttons. The system clock at the bottom right indicates 4:45 PM on 2/2/2025.

Figure 5
12.6.1 Creating a remote access VPN connection



Figures 1 through 5 show that all the lab questions were answered correctly.